

MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL-1902



Department of Information and Communication Technology

LAB Report

Lab Report No: 02

Lab Report Title: Encapsulation and Data Security in Java:
BankAccount Example.

Course Title: Software Engineering and Project Management Lab

Course Code: ICT-3108

Submitted By-	Submitted To-
Naeema Fairouz ID: IT-23048 3 rd Year 1 st Semester Session: 2022-2023 Dept. of ICT, MBSTU	Dr. Ziaur Rahman Professor, Dept. of ICT, MBSTU

Date of Performance:

Date of Submission:

Lab Title : Encapsulation in Java.

Objective : To understand encapsulation using private variables and methods.

Theory :

Encapsulation hides data using private variables. Data is accessed through public methods with validation. This improves data security and integrity.

Program Code :

```
class BankAccount {  
    private String accountNumber ;  
    private double balance ;  
  
    public void setAccountNumber (String acc) {  
        if (acc != null && !acc.isEmpty()) {  
            accountNumber = acc ;  
        }  
    }  
  
    public void setInitialBalance (double amt) {  
        if (amt >= 0) {  
            balance = amt ;  
        }  
    }  
  
    public void display() {  
        System.out.println (accountNumber) ;  
        System.out.println (balance) ;  
    }  
}
```

```
public static void main (String[] args) {  
    BankAccount b = new BankAccount ();  
    b.setAccountNumber ("ABC123");  
    b.setInitialBalance (5000);  
    b.display ();  
}  
}
```

Output :

ABC123

5000.0

Conclusion :

Encapsulation protects data from invalid access. Validation ensures correct data entry.